GREENSTAR NATIONAL COMPUTER CLUB
(BASED IN THE VAAL-TRIANGLE)

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CLUB MONTHLY NEWSLETTER

30th NOV '86

BELTEL

HARDWARE BARGAINS

UNDERSTANDING COMPUTERS

MSX

The great Christmas extravaganza

SVI-328

ALL THE CHRISTMAS SOFTWARE

Flight Path 737

CHRISTMAS-HARDWARE-BARGAINS

THE FOLLOWING ITEMS ARE AVAILABLE ONLY WHILE STOCKS LAST.

SVI.738..... 'XPRESS MSX COMPUTER _____ R 1000

SVI.838..... 'X-PRESS 16 MS-DOS COMPUTER _____ R 2000
(No monitor)-256kB RAM
1 x D/S-D/D DISK-DRIVE

SVI.328..... GREAT-PACK: 1 x 328 MkII;
1 x 904 Datacorder;
ONLY 10 LEFT 1 x 101 Joystick;
1 x 105 Graphic Tablet ___ R 800

LLOYDS COLOUR MONITOR / MED. RES _____ R 900

GST and POSTAGE INCLUDED

MAXI-EXPANDER FOR SVI.728 _____ (6 LEFT) ____ R 60
SPEECH SYNTHESIZER _____ (10 LEFT) ____ R 90
SOFTSWITCH & AUDIO-AMPLIFIER _____ (4 LEFT) ____ R 70
DAISY ADAPTER FOR MSX COMPUTERS ____ (15 LEFT) ____ R 20
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@@@ THE ABOVE DAISY SYSTEMS HARDWARE IS THE LAST,
ACT NOW....DAISY WILL NO LONGER PRODUCE THEM @@@

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Market Leaders CLUB PRICE R 1000

Then along came . . . EXPRESS!—

She looks just what she is . . . a market leader

Just when everyone thought they were catching up, Spectravideo has launched 'Express", thus maintaining their place as market leaders.

Spectravideo set the pace for home computers all over the world when it developed the concept of MSX. Until then the home computer industry had no direction and manufacturers went their own and, in most cases, "not so sweet" way.

Standardisation was a pipe dream and was confined to the same conversations in which the possibility of life on Mars was discussed. Standardisation was a nice concept for home computers but never something practical and hardly possible.

But then came Spectravideo, and its revolutionary MSX operating system. Suddenly it was possible for peripherals and software to be interchanged irrespective of the make of computer.

Suddenly it wasn't so strange anymore, after all interchangeability in the hi-fi business had been a watch word ever since Pa fell off the bus. Other manufacturers rushed to sign the contract allowing them to embrace the MSX operating system.

It was natural they should. Suddenly there were about 20 of the world's major computer manufacturers committed to producing MSX machines. The consumer had an awesome choice of hardware and software. New MSX machines began arriving on the market with almost monotonous regularity, all of them trying to emulate the success of the Spectavideo 728 MSX computer.

But just when they think they're catching up what does Spectravideo go and do?

It launches Express, a dynamic new direction in MSX computers. The SVI Express is an enhanced MSX computer system capable of performing useful applications. In fact all the hardware is intergrated into one ergonomically-designed console. Everything you need comes ready built in. In essence the Express is a 'drive home' machine.

In addition to the standard MSX features the Express includes:

- ★ Built-in 3½" micro floppy disk drive;
- ★ Full-stroke keyboard with 73 keys;
- ★ 80K Ram (ie. 64K user Ram and 16K video RAM);
- ★ Two general purpose I/O ports;
- ★ Enlarged cursor control keys;
- ★ Multi-purpose retractable handle.

The Spectravideo Express is more than just a games machine. It is a versatile computer with many real-world applications. It can run MSX DiskBasic, MSX-DOS and CP/M software.

Express can also be connected to any RS232C or Centronics Parallel device.

But as if that was not enough the Spectravideo Express is supplied with an inte-grated 4-in-1 software package that includes a memo-writer, spreadsheet, file handler and report writer.

With these programs Express will allow the end user to produce flawless memos, organise business information, perform spreadsheet calculations, prepare professional reports and generate high resolution graphs and charts.

The Spectravideo Express is not the type of machine one will want to leave at home. It at last makes a home computer a useful tool.

ARTIFICIAL INTELLIGENCE

The Costa del Sol's voluptuous sun and rhythmic surf took their toll on Geddy's ability to focus on her budding business ventures, and eventually she tired of Roger's youthful enthusiasms as well. Randless, unwelcome at home and in her native Wales, Geddy fled the old world for the new. As have many desperados and high flyers before her, Geddy arrived in the land of middle-class hustlers and anything-goes free enterprisers looking to seek her fame and fortune – and my forgiveness. Geddy was in America.

I was well aware of this when, early one Monday morning in late May, Geddy appeared on my doorstep like a Jehovah's Witness from the church of high tech. She had arrived the previous day, and had hitched a ride from Logan airport with a Dec systems engineer to one of the techies' favourite watering holes along America's technology highway, where she mingled and flirted with some of America's best and brightest. All the while she was looking for the answer or at least a good angle, when suddenly she realized that here, anything indeed was possible.

All that remained was to fetch her kids from Johannesburg, to set up shop in my attic – oh, and of course, to come up with the next 1-2-3.

G: "The market is chock-a-block with word processors. Desktop organizers are dog-eat-dog. General ledger is generally boring. Utilities are developed by nerds who get their jollies looking at disk sectors. Idea processors are an idea who's time hasn't yet come. And spreadsheets, well, you know how I feel about Symphony, luv – who can improve on perfection? Graphics suffer from a lack of standards and clear direction. DBMS is arcane and getting arcaner by the day. And what can one say about communications? What's left, babes? I need a winner."

Pondering her dilemma for a long moment, distracted by her off-the-shoulder Spanish sun dress just barely covering her stunning tan, I took the easy road.

CC: "Let's consult the computer."

G: "Listen Cline, I'm no dolly bird. I know computer's can't think."

How could she say that? Hadn't she heard that both Mitch Kapor and Bill Gates have said that artificial intelligence is the next great growth area of personal computers? Didn't she know that both Lisp and Prolog, favorites of the AI crowd, were now available in a dozen-odd versions for the IBM PC & clones? Weren't new expert-system and decision-making packages, such as Expert System's Expert-Ease and Lightyear's Lightyear, being ported to or developed for pcs almost daily? And what of Lotus's funding of Arity, founded by two renegade Lotus employees, dedicated to making software 'smart?'

G: "Rubbish. Show me."

We popped Lightyear into my pc, and began working through the three-step process required to reach a decision. The first step was to identify the alternatives. This presented a bit of a problem, since Geddy's principle alternative was to get rich, the only alternative to that being its opposite (to someone bitten with the Route 128 gold rush bug, "getting a little rich" isn't a meaningful selection).

We settled on identifying generic categories and/or vertical markets within which she was most likely, given her skill mix, capital base, and sexual preferences, to get rich, and then proceeded to identify the criteria the alternatives should be judged on (including "My dream product should utilise virtual memory and be sexy"), as well as the 'verbal' intangibles that Lightyear allows one to include, such as its 'risk' vocabulary consisting of the words 'safe', 'minimal', 'chancy' and 'dangerous'. After adding a final category of guidelines or 'rules' relevant to the decision (in which I insisted that "It must be legal" be included), it was time to have the program 'evaluate' the decision.

The computer screen blanked, and was replaced a moment later by a simple screen message: "Take two aspirin and go to bed."

Geddy thought it over, then looked up at me: "Wrong pill, but right idea. Let's go, Babes – doctor's orders."

I stammered that something had gone wrong, that somehow her data file must have gotten chained into a medical expert decision system I was devising.

Geddy interrupted me, her eyes flashing: "I'm more likely to believe computers can think than that lame rationalization. No matter, luv, come hither . . . ."

CC: "But that's the wrong answer."

Geddy, yawning, beckoning: "Don't argue with superior intelligence, Mr C."

Of course, Mitch Kapor was *also* quoted at a recent pc conference as having said: "Artificial intelligence will become the most despised and misused buzzword in the next year." He foresaw that there would be a "lemming-like rush" to AI that will be as disastrous as the integrated software bandwagon that many developers jumped onto (and subsequently fell off of) last year. So which is it, Mitch? You can't have it both ways.

Why such a fuss about something as esoteric and frequently misunderstood as AI? Partly because we all have grown up *expecting* computers to think. Admit it, you also had to hold back tears when Hal was lobotomized in the film *2001*. And the more recent *Electric Dreams* hypothosized a computer who could acutally fall in love with its owner's lover. Damned seductive concept.

The big boys have been beavering away on the concepts and problems of AI ever since computers were invented, with key concepts of pioneers such as Marvin Minsky (one founding father of MIT's famed AI lab) beginning to see practical use in some of the expert systems available today. Expert-system programs absorb technical knowledge from human specialists, apply it to varied situations, and reach conclusions. Some of the best expert systems available today in mainframeland function well because of the 'closed' nature of the fact sets (such as geological data or crime leads) they use. In a well-defined fact set, or universe, cross-correlation of disparate but similar facts is rather straightforward, at least in computer terms. Beyond these systems, which are only beginning to find useful application in the micro world, there are the more ambitious projects such as Japan's fifth generation (for which they've mobilized every yen & yang of Japan Inc.), and even more bizarre developments in the area of 'molecular electronics' (in which the basic building blocks of life do double duty as the basic circuitry of advanced electronics – at which point who's to say what will be master and who will be slave – Marvin Minsky reportedly once half-joked, "If we're lucky, perhaps they (computers) will keep us as their pets").

—HOW SMART IS IT?

Well, we in the micro world don't have to concern ourselves with such lofty issues quite yet, 'cause in spite of the latest advances, the really scary stuff in AI is still beyond the ken of today's pcs. No one seriously suggests any longer that an intelligent computer can be realized purely algorithmically. Rather, the process will be a long and painstaking one that involves cramming more and more facts into the machine. All the AI research done to date suggests that if computers are to begin to approach the capabilities of the human brain, they must store and retrieve huge amounts of context data, quickly. Very quickly. Which is why the latest AI machines use hundereds of 68020 and/or bit-slice screamers ganged together specifically to facilitate parallel processing of huge amounts of data.

Expert and decision-making systems are the first truly useful AI applications squeezed into a pc. Besides the two mentioned earlier, which were expressly developed for pcs, there are several expert systems that have been downsized from mainframe counterparts. For example, Reveal is a $4 500 package that was cloned from Infotym's $50 000 mainframe package. The Reveal development environment consists of three major components: a table of matrix data, a program model describing the relationships among the matrix rows and columns, and, optionally, a 'policy' program that can contain any relevant knowledge-based rules that can be called by the program model.

This last component points to one of the key elements of any expert system – you have to know what you're talking about to set up a system effectively. Pc-based expert and decision-making systems require *you* to supply the contextual data and decision criteria for a rather narrowly defined universe of tasks, which is what makes the entire proposition of pc-based AI feasible. In other words, an expert system is only as good as the experts who contribute the inference procedures and knowledge-based rules to the system. You can't sit down to build an expert system on, say, quantum mechanics, unless you know something about the subject.

This may seem self-evident. But I think what Kapor was trying to say in his second quote is that AI will be abused, just as the integrated package market was earlier, by those who jump on the bandwagon without understanding AI's limitations in pc applications (confusing, perhaps, the reality with the fiction in their own imaginations?) – that a machine is still, and will continue to be for quite some time, only as smart as those who are developing the software, and those using it.

All of which is why Lotus and Arity may have the best idea for applying these concepts to pcs – by marrying AI concepts to word processing, electronic spreadsheets, and financial analysis, personal computers will be far easier to use, and more accessible to all of us. In this context AI is not a replacement for human expertise, but rather a means to make computers less user-surly when it comes to doing the truly boring, mundane tasks that frustrate even the most slavish computerphile.

Sort of an electronic manservant, a Jeeves. Hal in a laptop will have to wait – although this won't necessarily stop Geddy (and Kapor, and hundreds others like them) from making their fortune riding this wave.

CAD/CAM

Good news for geologists, cartographers, facilities planners, artists, and computer designers – computer-aided design (CAD) software is plummeting in price, moving from the heady realm of the expensive mini or mainframe installation onto the most humble of personal computers. And the hard times that have fallen on most of the grown-up side of this industry – Computervision recently retrenched 950 people worldwide after incurring a loss on its last quarter operations – has curiously (perhaps portentiously) not fallen on the smaller CAD/CAM manufacturers who have aggressively incorporated microcomputers into their product lines.

Autodesk's Autocad, the first inexpensive CAD product to appear widely in retail computer stores, is currently

the hottest pc CAD product. Priced at $1-2 000 depending on the number of advanced options selected, it works on a wide variety of 16-bit micros. Autocad appeals to a wide range of users, as does Drawing Processor 2, a $995 package from BG Graphics.

For somewhat higher prices you can obtain micro-based CAD packages for more specific applications. Personal CAD's CADplan, meant for use mainly in architectural engineering and facilities planning, has sold more than 5 000 copies, and Celestial Software offers programs aimed at mechanical engineers. Some 20 companies offer computer-aided manufacturing (CAM) products for micros, although most are still priced in the $10 000 range.

Industry analysts predict prices will fall even further, to the $200 to $1 000 range. "It will take CAD out of the architectural and engineering level and make it available to every art student or high school student in the country," said one industry insider. However, Charles Foundyller, president of Daratech, a Cambridge, MA. CAD research firm, warns "The big boys are coming." As in any growth industry, it won't take long for the IBM's and Computervisions to recover from their mistakes and to come out with micro-based CAD products of their own.

Softcon

Play a dirge for yet another industry trade show – Softcon '85 in Atlanta was an unmitigated bore by all reports. Most of the software heavyweights, such as Lotus, Ashton-Tate, Microsoft, Borland, Microrim, Multimate, and Micropro gave it a miss. "As far as I'm concerned, you can kiss this show goodbye," said one disappointed vendor.

I would normally attempt to describe a bit of what did happen at the show, but there really wasn't enough of sufficient interest to waste prime verbiage on it. However, I was amused to hear that one vendor, Gold Nugget Jewelers of Atlanta, was doing a brisk business in a booth near the back of the show. Organizers were reportedly at a loss to explain how a jewelery wholesaler secured booth space at a software show. ■

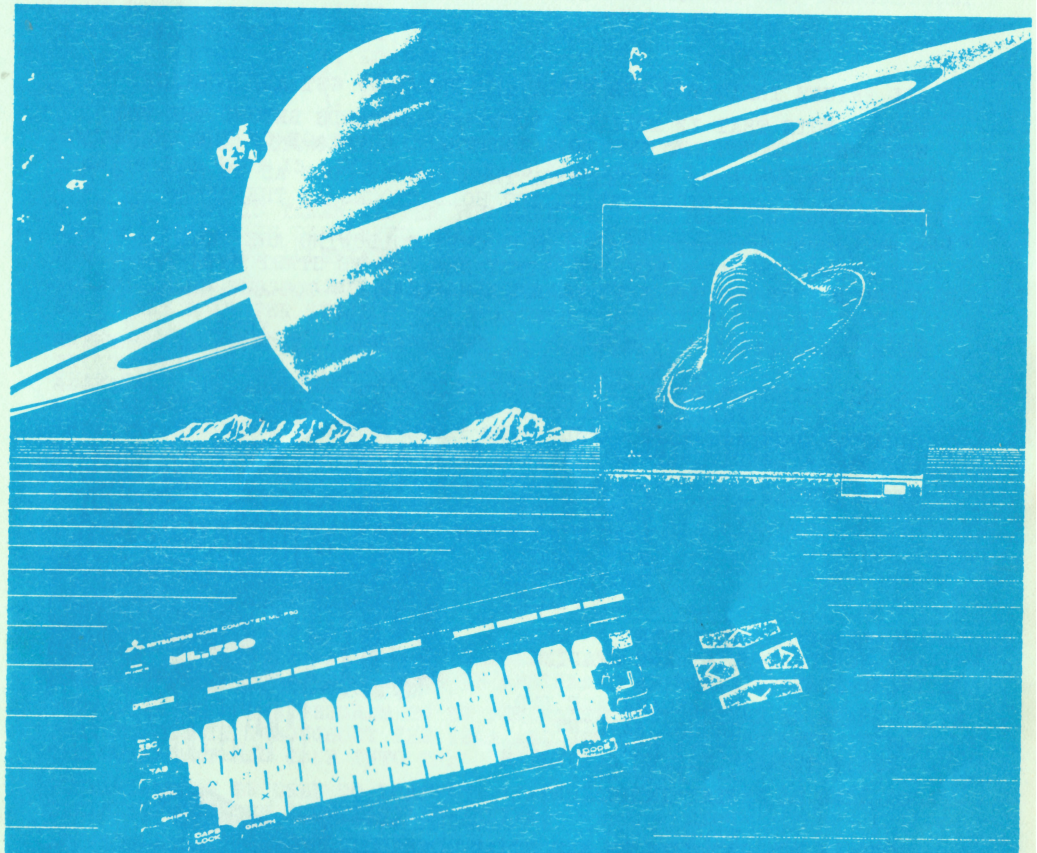
MITSUBISHI
MSX COMPUTER R 600 incl.

Specifications

CPU	Equivalent to Z80A (3.6MHz)
Memory	
ROM	BASIC 32KB utility program (FX2) 16KB
RAM	Main 64KB V-RAM 16KB
Screen display	
Text mode	32-character, 24-line display
Graphics mode	256 dots ×192 dots
Colour display	16-colour sprite function
Output method	RF output, composite video output
Keyboard	
Keyboard layout	QWERTY English
	Alphanumeric, special character
	Control and special keys
	Cursor control keys
	5-function keys (10 functions using shift)
	Total of 73 keys
Format	Integral with CPU
Sound function	8 octaves and sound effects
	Three tone harmonies
	Adjustable volume
	External output terminals attached
Cassette interface	Internal transfer speed: 1200 or 2400 baud, remote controllable
Printer interface	Internal Centronix® specification standards
ROM cartridge	Plug-in (two slots)
Connectors and terminals for peripheral devices	
	Joy-stick connector sockets: 2 (9-pin)
	ROM cartridge connector sockets: 2 (50-pin)
	Printer connector sockets: 1 (14-pin)
	RF output jack: 1 (RCA phone)
	Composite video output jack: 1 (RCA phone)
	Cassette I/O connector socket: 1 (8-pin DIN)
	Audio output jack: 1 (RCA phone)
External Dimensions (W × H × D)	424 × 230 × 60mm

MITSUBISHI
MSX COMPUTER SYSTEMS

ML-F80



A Computer for Today—and All Your Tomorrows...

Much more than a superb game computer, the ML-F80, ML-FX1 and ML-FX2 offer the whole family entry to the home-computer age. As simple to operate as a push-button telephone, and connecting directly to your colour TV, they are designed for home use, putting the full power of the computer at your service. Put it to work at cheque balancing and household accounts, storing names, addresses, phone numbers, recipes, record collections, diets, appointments... It's a modern magic genie that helps keep the home running smoothly.

TV picture simulated

XMAS...XMAS...XMAS...SOFTWARE REVIEW...XMAS...XMAS....XMAS

SPECIAL SOFTWARE REVIEW FOR XMAS 1986

ALL PRICES INDICATED ARE FULLY INCLUSIVE OF GST AND POSTAGE.

PLEASE NOTE THAT IF DELIVERY SOMETIMES EXTENDS BEYOND THE TWO WEEK PERIOD, THIS IS DUE TO A TEMPORARY SHORTAGE OF THE PROGRAM/S.

PLEASE BE PATIENT; WE DESPATCH PROGRAMS AS SOON AS WE RECEIVE THEM FROM THE DISTRIBUTORS. WE DO NOT HOLD STOCKS AT THE CLUB.

SOME OF THE SOFTWARE (TAPES OR BOOKS) FOLLOWING ARE PAST OR WELLKNOWN BUT SOME ARE DEFINITELY NEW AND WE START WITH THE PROGRAM THAT ALL OF YOU HAVE BEEN WAITING FOR.....

FLIGHT PATH 737.....328/MSX.....SIMULATOR.....TOPS.....R 25

AS THE PILOT OF THIS HIGH PERFORMANCE JET-AIRLINER, YOU MUST TAKE OFF FROM AN AIRFIELD SURROUNDED BY HIGH MOUNTAINS AND HAVING CLIMBED YOUR AIRCRAFT SAFELY OVER THEM, PREPARE YOURSELF AND THE JET FOR A LANDING AT AN AIRFIELD IN THE VALLEYS BELOW.

SKILL LEVELS..... FIRST SOLO
STUNT PILOT
PART-TIME PILOT
EXPERIENCED
PROFESSIONAL
TEST PILOT

CONTROLS.....ALTITUDE
AIRCRAFT HEADING
AIRSPEED
UNDERCARRIAGE IND.
GROUND WARNING
ARTIFICIAL HORIZON
AIRSPEED IND.
FLAP INDICATOR
FUEL
TIME
RUNWAY HEADING
DISTANCE TO DEST.
RATE OF CLIMB
RATE OF DESCENT
APPROACH HI/LO IND.

SCREEN MESSAGES..... 39

GRAPHICS.....FULL ROTATIONAL

NOW FOLLOWS 50 MORE REVIEWS OF THE TOP PROGRAMS AND BOOKS.....

R15

FUNMASTER..... 328/MSX / EDUCATIONAL / NEW RELEASE / R 25

FUNMASTER IS A PROGRAM DESIGNED TO ASSIST STUDENTS IN THE PREPARATION FOR EXAMINATIONS. YOU ONLY NEED TO PURCHASE FUNMASTER ONCE. YOU WILL BE ABLE TO LOAD ANY 'FUN TO LEARN' SUBJECT FILES (FOR ANY STANDARD). ALL INSTRUCTIONS ARE 'ON-SCREEN' AND THE PROGRAM IS VERY USERFRIENDLY EVEN FOR SEVEN-YEAR-OLDS'.
THE MENU IS SIMPLE.....

1. LOAD A FILE
2. VIEW QUESTIONS & ANSWERS
3. TAKE THE TEST
4. TERMINATE

THE FILES LOAD AUTOMATICALLY WITHOUT THE NEED FOR 'NAMES' ETC.
VIEWING THE TEST IS LIKE REVISION AS THE ANSWERS ARE DISPLAYED WITH THE QUESTIONS.
WHEN THE TEST IS TAKEN, AFTER YOU MAKE YOUR CHOICE; THE SCREEN CLEARS AND THE ANSWER IS DISPLAYED. THIS QUESTION BY QUESTION RESPONSE HELPS YOU REMEMBER AND ALLEVIATES THE NEED FOR 'SCORING'.

FUN-TO-LEARN SUBJECT FILES ARE AVAILABLE IN A VARIETY OF SUBJECTS AND COST ONLY R 15 PER SUBJECT PER STANDARD. HERE IS A LIST OF THEM.....

MATHEMATICS...GRD.2 / STD.1 /2 /3 /4 /A5 /B5
TRIGONOMETRY.....STD.8
ALGEBRA.....STD.8 /9-10
GENERAL SCIENCE.....STD.2 /3 /4
PHYSICS.....STD.5
BIOLOGY.....STD.5
APPLIED MATHEMATICS.....STD.9-10
MECHANICS.....STD.9-10
ELECTRICITY.....STD.9-10
CHEMISTRY.....STD.9-10
BOTANY.....STD.9-10
GENETICS & ECOLOGY.....STD.9-10
METABOLISM.....STD.9-10

THE LIST IS BEING EXPANDED (ON REQUEST) AND THE PRIMARY-SCHOOL SUBJECTS ARE SOON TO BE AVAILABLE IN AFRIKAANS AS WELL. (PLEASE ENQUIRE)

FUNMASTER IS NOT A PROGRAM TO BE PURCHASED JUST BEFORE AN EXAM. IT IS BEST TO USE IT THROUGHOUT THE YEAR AND GET THE FULL BENEFIT OF ALL THE WORK THAT HAS GONE INTO THE COMPILATION OF THE SUBJECT-FILES.

FUNMASTER CAN BE USED TO LOAD ANY 'TEACHERS-PET' FILES AS WELL AS ALL FILES SUPPLIED ON THE EXAMINER-SERIES.

AS WITH ANY GREENSTAR PROGRAMS, TAPES CAN BE REPLACED FOR R5 ONCE THEY BECOME WORN OR DAMAGED.

DON'T WAIT FOR A BAD REPORT CARD. START NOW.

XMAS SPECIAL R12

MICRO - JAM..... 328 ONLY / ARCADE GAME / NEW RELEASE /R 20

SOMETHING IS WRONG WITH YOUR COMPUTER CIRCUITS. YOU HAVE DESIGNED A ROBOT THAT HAS BEEN MINIATURISED AND PUT INTO THE FAULTY MICRO-CHIP. THE PROBLEM IS THAT THE MICRO-CHIP HAS BEEN INVADED BY BUGS AND LOTS OF OTHER NASTIES WHICH ARE DESTROYING THE DATA. YOUR TASK IS TO TRY AND SAVE AS MUCH OF THE DATA AS POSSIBLE. THERE ARE TWENTY BITS OF DATA IN EACH SCREEN. YOU MUST GET TO EACH DATA-BIT FIRST (IF YOU CAN)

VESTRON.....328/MSX / ARCADE / NEW RELEASE / R 20

THE YEAR IS 2024. EARTH IS DYING. THE AMSTRADIAN EMPIRE HAS TAKEN OVER THE UNIVERSE AND EARTH IS ONE OF THE LAST REBEL STRONGHOLDS - BUT YEARS OF FIGHTING HAVE DEPLETED EARTH'S ENERGY. HOWEVER, SCIENTISTS HAVE DISCOVERED A NEW ENERGY FORM CALLED CALOSCIN - WHICH WHEN MIXED WITH WATER CAN BE USED AS A SUBSTITUTE FOR OIL. VESTRON, AN OLD IRON-ORE MINING PLANET, IS THE ONLY PLANET ON WHICH THIS SUBSTANCE CAN BE FOUND, IN THE FORM OF CRYSTALS.
AMSTRADIANS HAVE NOT YET DISCOVERED THE SIGNIFICANCE OF THE CRYSTALS BUT NEVERTHELESS, KEEP THEM HEAVILY GUARDED. THE LARGEST CRYSTAL IS SITUATED IN A VAULT, WEST OF THE MAIN MINE WORKINGS. IT IS YOUR TASK TO FIND THAT CRYSTAL, IF YOU CAN.

FOOT VOLLEY.....328/MSX / ARCADE / NEW RELEASE / R 20

AS THE OPENING APPLAUSE DIES, THE CLOCK BEGINS. THIS IS YOUR CHANCE TO PROVE THAT YOU ARE A GREAT FOOTBALLER, IN THIS TEST OF AGILITY & VOLLEY SKILL. ONCE COMMENCED, YOU THE PLAYER CAN CHOOSE TO BE ONE OF THREE PROFESSIONAL FOOTVOLLEY PLAYERS.
YOU MUST JUGGLE THE BALL AND SCORE POINTS BY KICKING AND RETRIEVING, USING DIFFERENT COMBINATIONS OF KICKS.
MISTAKES COST TIME AND POINTS. SKILL IS REWARDED.

KNIGHT TYME.....328 ONLY / ARCADE / TOP PROGRAM / R 25

I LANDED HEAVILY ON THE TRANSPORTER PAD OF THE USS PISCES AND WHILE I WAS BENDING OUT SOME OF THE DENTS IN MY ARMOUR, I WAS APPROACHED BY A STRANGE METAL CREATURE. HE GAVE ME A CUBE-LIKE OBJECT WHICH I LATER LEARNED WAS A DATA-CUBE. AS I HELD IT I BECAME AWARE OF A STRANGE POWER INSIDE ME. AS THE DATA-CUBE TRANSFERRED ITS DATA TO MY BRAIN-CELLS, I STARTED TO FORMULATE MY PLAN FOR SURVIVAL.
I WILL HAVE TO LOCATE THE TYME GUARDIANS, IF THEY STILL EXIST, AND OBTAIN A TYME-MACHINE FROM THEM. MY FIRST PRIORITY IS TO.....

SPACE WALK.....328 ONLY / ARCADE / TOP PROGRAM / R 20

YOU ARE AN ASTRONAUT IN COMMAND OF A SPACE SHUTTLE. FROM YOUR BASE ON THE MOON YOU KEEP WATCH FOR STRAY SATELLITES AND TRY TO RETRIEVE THEM. BATTLE YOUR WAY ACROSS THE LUNAR SURFACE THEN USE YOUR JET-PACK TO THRUST UP TO THE SATELLITE. RENDEZVOUS WITH THE SATELLITE BEFORE IT CRASHES INTO THE LUNAR LANDSCAPE - THEN CARRY IT BACK TO YOUR LAUNCH-PAD, WHERE YOUR SPACE-SHUTTLE IS STATIONED. AVOID THE ASTEROIDS, ALIEN COSMONAUTS, MALFUNCTIONING ROBO-DESTRUCTORS, LANDERS LUNAR PODS, ROBOTS, ETC. 16 DIFFERENT LEVELS OF ICE COLD, LOW PRESSURE DEATH AWAITS YOU.

SPEED KING.....328/MSX / ARCADE / TOP PROGRAM / 328: R 25
MSX: R 30

THE MOTORCYCLE RACING GAME WITH THRILLING HIGH-SPEED WHEEL-TO-WHEEL ACTION AGAINST 19 OTHER DETERMINED RIDERS. WORK YOUR WAY UP TO WORLD CLASS WITH 3 SKILL LEVELS ON 10 WORLD-FAMOUS TRACKS.

MANIC MINER.....328/MSX / ARCADE / TOP PROGRAM / R 25

MINER WILLY, WHILE PROSPECTING DOWN SURBITON WAY, STUMBLES UPON AN ANCIENT LONG-FORGOTTEN MINE-SHAFT. ON FURTHER EXPLORATION, HE FINDS EVIDENCE OF A LOST CIVILISATION FAR SUPERIOR TO OUR OWN, WHICH USED AUTOMATONS TO DIG DEEP INTO THE EARTH'S CORE TO SUPPLY THE ESSENTIAL RAW MATERIALS FOR THEIR ADVANCED INDUSTRY. AFTER CENTURIES OF PEACE AND PROSPERITY, THE CIVILISATION WAS TORN APART BY WAR AND LAPSED INTO A LONG, DARK AGE, ABANDONING THEIR INDUSTRY AND MACHINES.
NOBODY, HOWEVER TOLD THE ROBOTS TO STOP THEIR MINE-WORKING AND THROUGH COUNTLESS AEONS THEY HAD STEADILY ACCUMULATED A HUGE STOCK-PILE OF VALUABLE METALS AND MINERALS. MINER WILLY NOW REALISES THAT HE HAS THE CHANCE TO MAKE HIS FORTUNE BY FINDING THE UNDERGROUND STORE. HOW HE DOES THAT DEPENDS ON YOU AND.....

FINDERS KEEPERS.....328/MSX / ARCADE / TOP PROGRAM / R 20

THE KING OF ISBISIMA IS, TO SAY THE LEAST, A LITTLE UPSET.
TOMORROW IS HIS DAUGHTER'S BIRTHDAY AND HE HAS NO IDEA WHAT TO GET HER. AS MAGIC KNIGHT YOU HAVE BEEN ORDERED TO FIND THE PRINCESS GERMINTRUDE A VERY SPECIAL PRESENT. THIS MAY BE YOUR CHANCE TO PROVE THAT YOU ARE WORTHY OF JOINING THE FAMED 'POLYGON-TABLE', THE HIGHEST HONOUR ANY KNIGHT COULD WISH FOR.
THE KING TRANSPORTS YOU TO THE CASTLE OF SPRITELAND WHICH IS TEEMING WITH MANY WIERD AND WONDERFUL CREATURES. ALSO THERE ARE TWO MAZES WHICH CONTAIN MANY OF THE QUEEN'S RELATIVES. APART FROM THE GRISLY GHOULS AND ENERGY-SAPPING CREATURES THERE ARE GHOSTLY TRADERS WHO ARE OFTEN QUITE WILLING TO EXCHANGE THE TREASURES THAT YOU FIND IN THE CASTLE FOR MONEY WHICH THEY STOLE IN THEIR FORMER LIVES.
SOME OBJECTS WILL REACT WITH EACH OTHER TO FORM EITHER MORE OR LESS VALUABLE OBJECTS, TO SELL OR COLLECT. YOU WILL BE TEMPTED.....

SPACE FORCE.....328 ONLY / ARCADE / SD685 / R 15

FAR AWAY IN BOUNDLESS OUTER-SPACE, AN EVIL POWER; THE 'BLACK STAR EMPIRE' ; IS GROWING AND EVER-EXPANDING, THREATENING THE PEACE OF THE UNIVERSE. COMMISSIONED BY THE GREAT COMMITTEE, THE 5 STAR-FIGHTER SPACESHIPS, EQUIPPED WITH THE MOST ADVANCED HIGH-QUANTUM ENERGY EMITTER, ARE GOING TO DESTROY THIS FORMIDABLE FORCE.
WHO WILL WIN THE BATTLE? THE WICKED OR THE RIGHTEOUS? THAT ALL DEPENDS ON YOU, THE STAR FIGHTER CAPTAIN. 16 LEVELS AND UP TO 40 ENEMIES PER LEVEL.

CHILLER.....328/MSX / ARCADE / TOP PROGRAM / R 20

IT IS A NIGHT FULL OF DARK FOREBODING. YOU ARE FRIGHTENED BOTH FOR YOURSELF AND FOR YOUR LOVED ONE, WHO IS IMPRISONED IN A HAUNTED HOUSE SOME DISTANCE AWAY. AS YOUR CAR APPROACHES THE FOREST, IT COUGHS TO A HALT AND THE ENGINE DIES.....

INTERNATIONAL KARATE...328/MSX / ARCADE / TOP PROGRAM / R 40

INTERNATIONAL KARATE IS A SIMULATION OF A KARATE TOURNAMENT WHERE TWO PEOPLE CAN COMPETE AGAINST EACH OTHER OR ONE CAN PLAY AGAINST THE COMPUTER. DEPENDING ON YOUR LEVEL OF SKILL YOU WILL FLY AROUND THE WORLD TO FIVE DIFFERENT LOCATIONS. YOU WILL ALSO HAVE TO COMPETE IN A SPECIAL BONUS ROUND.
AFTER EACH BOUT, YOU ARE GIVEN THE OPPORTUNITY TO SMASH A STACK OF TILES WITH YOUR FOOT AFTER WHICH YOU PASS TO THE NEXT LOCATION.

TETRA HORROR.....328 ONLY / ARCADE / TOP PROGRAM / R 20

THE ADVENTURER, TAMO, GOT A TREASURE MAP OF TETRA-CITY, WHICH EXISTED MANY THOUSANDS OF YEARS AGO IN THE MIDDLE OF SOUTH-AMERICA. ARMED WITH A FIRE-GUN WHICH HAS BEEN FUELLED IN THE FROG-STEPS, TAMO SETS OUT TO DESTROY ALL THE SKULLS, MONSTERS AND AVOID ALL THE ROLLING-STONES BEFORE HE SEIZES THE TREASURE. WATCH OUT. ONE MISTAKE ON THE NARROW PATH OR STAIR WILL CAUSE HIM TO FALL ONTO THE KNIFE-EMBEDDED GROUND.

ZAKIL WOOD.....MSX ONLY / ADVENTURE/ TOP PROGRAM / R 30

FROM THE DEPTHS OF ZAKIL WOOD YOU MUST COLLECT THE RUBY WHICH HAS PROTECTED YOUR VILLAGE FOR CENTURIES. IF YOU CAN PASS THE TWO-HEADED PYRAL THEN THE ADVENTURE STARTS IN EARNEST.
ZAKIL WOOD IS A COMPLETE ADVENTURE.

MAYHEM.....MSX ONLY / ARCADE / TOP PROGRAM / R 40

IN THESE THREE GAMES SCENARIOS YOU HAVE THE DAUNTING TASK OF RESCUING THE TRIBBLES FROM THEIR DYING PLANET AND PROTECTING THEM AS THEY CAUSE MAYHEM DURING THE JOURNEY BEFORE YOU CAN LAND THEM SAFELY ONTO THEIR NEW PLANET.
THIS IS AN ORIGINAL GAME WHICH FEATURES HECTIC MULTI-SPRITE ACTION.

CRAZY GOLF.....328/MSX / SIMULATION / TOPS / 328: R 40
MSX: R 30

CRAZY GOLF IS A FUN GAME THAT FEATURES UNIQUE GRAPHIC OBSTACLES WHICH MUST BE CIRCUMVENTED BEFORE THE BALL CAN BE PUTTED INTO THE HOLE.
THE PAR FOR EACH HOLE IS DISPLAYED AND A NOVEL DIRECTION-POINTER COMBINED WITH THE FORCE-INDICATOR ENABLES YOU TO TRY TO EQUAL OR BETTER PAR.
CRAZY GOLF IS CRAZY BUT FUN FOR EVERYONE.

HUMPHREY.....328/MSX / STRATEGY / TOPS / R 30

IN THIS GAME WE MEET HUMPHREY, THE LITTLE CHAP WHO LIKES TO KEEP ONE STEP AHEAD. WELL, THE FACT IS HE HAS TO, BECAUSE HE IS BEING FOLLOWED BY BOUNCING ATOMIC BOMBS.

JUSTWRITE JR.....328 CARTRIDGE / UTILITY / THE ONLY ONE / R 99

THIS IS AN EXTREMELY USER-FRIENDLY WORD-PROCESSOR DESIGNED SPECIFICALLY FOR LIGHT-DUTY. IT IS IN FACT, THE ONLY WORD-PRO FOR NON-DISK USERS. LUCKILY, IT IS A VERY GOOD ONE.

LOGO 1.....328/MSX / EDUCATIONAL / NEW RELEASE / R 25

THIS PROGRAM IS FOR BEGINNERS WHO WANT TO LEARN TO PROGRAM IN LOGO. THERE IS NO BOOK, BUT THE ACCOMPANYING PAMPHLET LISTS THE WORDS THAT ARE AVAILABLE, THEIR PARAMETERS AND A SHORT EXPLANATION. (ENG / AFR)

LOGO 2.....328/MSX / EDUCATIONAL / NEW RELEASE / R 25

THIS PROGRAM IS FOR MORE ADVANCED LOGO-USERS. THERE IS NO BOOK, BUT THE ACCOMPANYING PAMPHLET LISTS ALL THE WORDS, THEIR PARAMETERS AND GIVES A SHORT EXPLANATION. (ENG / AFR)

LOGO 1, THE BOOK.....328/MSX / EDUCATIONAL / FOR LOGO 1 / R 22

THIS BOOK IS AIMED AT THE VERY YOUNG USER AND EXPLAINS IN DETAIL, THE USE OF EACH WORD IN LOGO TOGETHER WITH PROGRAM EXAMPLES. THERE ARE LOTS OF FIGURES TO TRY AS WELL AS TWO FULL-LENGTH PROGRAMS.
IN THIS 60-PAGE BOOK, LOTS OF TIPS, CATCHES, PROCEDURES, INTRODUCTION TO CIRCLES, ANGLES, ETC. MAKE FOR A FULL-HOUSE OF FUN FOR THE LOGO NEWCOMER. AVAILABLE IN ENGLISH OR AFRIKAANS.

TEACHER'S PET.....328/MSX / EDUCATIONAL /ENG.OR AFR. / R 30

THIS IS THE MANAGEMENT PROGRAM FOR THE FUN-TO-LEARN SERIES AND IS USED TO CREATE THE SUBJECT FILES. THE PROGRAM CAN ALSO BE USED TO PREPARE EXAMINATION PAPERS USING THE LINE-PRINTER FACILITY.
AS WITH FUNMASTER, THE PROGRAM CAN ALSO BE USED TO TAKE THE TEST BUT IS NOT RECOMMENDED FOR YOUNGER USERS DUE TO THE COMPLEXITY OF USABLE FUNCTIONS. IT IS PASSWORD PROTECTED AND IS AVAILABLE IN BOTH ENGLISH OR AFRIKAANS.
IDEAL FOR TEACHERS OR STUDENTS FROM STD. 4 UPWARDS.

SPELLING TUTOR.....328/MSX / EDUCATIONAL / TOPS / R 20

THIS TUTOR CAN HANDLE WORDS OF ANY LENGTH IN FILES CONSISTING OF 10 WORDS AT A TIME WHICH CAN BE SAVED TO AND LOADED FROM TAPE AT WILL. YOU CAN INPUT YOUR OWN WORDS AND DRILLING IS DONE FIRSTLY BY THE WORD BEING REVERSED, VOWELS BEING LEFT OUT AND THEN LASTLY, BEING COMPLETELY MIS-SPELT. GOOD FOR ADULTS TOO.

BABY-DRAW-LOGO.....328/MSX / EDUCATIONAL / PRE-SCHOOL / R 20

THIS PROGRAM IS SPECIFICALLY DESIGNED FOR CHILDREN WHO HAVE NOT YET LEARNT TO READ. DRAWING IS SIMPLE AND CAN BE DEMONSTRATED INITIALLY BY THE PARENT, AFTER WHICH THE CHILD CAN BE LEFT TO HIS/HER OWN DEVICES. EVERY FUNCTION IS ICON-DRIVEN AND ALL THAT IS REQUIRED IS THAT THE CHILD KNOW HOW TO COUNT TO TEN.

GHOST-PANIC.....328 ONLY / ARCADE / TOP PROGRAM / R 20

YOU ARE IN AN ANCIENT CAVERN WHICH IS BEING HAUNTED BY GHOSTS. TREASURE HAS BEEN REMOVED FROM A TOMB AND SPREAD ABOUT THE CAVERN. YOU MUST COLLECT ALL THE TREASURE AND THEN PROCEED TO THE TOP OF THE TOMB, SITUATED BEHIND THE GREEN DOOR. ONCE YOU HAVE ACHIEVED THIS YOU WILL PROCEED TO THE NEXT SCREEN WHICH IS DIFFERENT AND MORE DIFFICULT.

R12
XMAS SPECIAL

JET SET WILLY.....328/MSX / ARCADE / TOP PROGRAM / R 25

MINER WILLY, INTREPID EXPLORER AND NOUVEAU-RICHE SOCIALITE HAS BEEN REAPING THE BENEFITS OF HIS FORTUNATE DISCOVERY IN SURBITON. HE HAS A YACHT, A CLIFF-TOP MANSION, AN ITALIAN HOUSE-KEEPER, A FRENCH-COOK AND HUNDREDS OF NEW-FOUND FRIENDS WHO REALLY KNOW HOW TO ENJOY THEMSELVES AT A PARTY. HIS HOUSE-KEEPER, MARIA, HOWEVER, TAKES A VERY DIM VIEW OF ALL HIS REVELLRY AND FINALLY, AFTER A PARTICULARLY BOISTEROUS THRASH, PUTS HER FOOT DOWN. WHEN THE LAST OF THE LOUTS DISAPPEARS DOWN THE DRIVE IN HIS ASTON-MARTIN, ALL WILLY CAN THINK ABOUT IS CRASHING OUT IN HIS FOUR-POSTER. MARIA, HOWEVER, WONT LET HIM INTO HIS ROOM UNTIL ALL THE DISCARDED GLASSES AND BOTTLES HAVE BEEN CLEARED AWAY. CAN YOU HELP WILLY OUT OF HIS DILEMMA? HE HASN'T EXPLORED HIS MANSION PROPERLY YET (IT IS A LARGE PLACE AND HE HAS BEEN VERY BUSY) AND THERE ARE SOME VERY STRANGE THINGS GOING ON, SOME VERY STRANGE THINGS INDEED.

MOLECULE MAN.....328/MSX / ARCADE / TOP PROGRAM / R 20

THE RADIATION GNAWS AWAY AT YOU, DRAINING YOUR PRECIOUS ENERGY RESOURCES - SAPPING YOUR LIFE-FORCE. THE MULTI-PERSPECTIVE 3-D MAZE IS HIDING THE VERY LIFE-PILLS THAT CAN BRING YOUR SALVATION. YOU ARE THE MOLECULE-MAN. YOUR PURPOSE IS TO SURVIVE.
LOST IN THE MIDDLE OF THE MAZE WITH TIME AND RADIATION AS YOUR ENEMIES, ESCAPE VIA THE TELEPORTER WOULD BE EASY BUT FOR THE FACT THAT IT CAN ONLY BE ACTIVATED BY THE INSERTION OF 16 CIRCUITS. THESE MUST BE FOUND AND COLLECTED FIRST. THERE IS MUCH MORE THAT YOU DO NOT KNOW. YOU WILL ONLY FIND OUT WHEN YOU PLAY THE GAME.

FORMULA 1 SIMULATOR...328/MSX / SIMULATION / TOPS / R 20

10 WORLD-FAMOUS TRACKS SET THE PACE FOR THIS ACCELERATION, BRAKING AND ROAD-HOLDING, REALISTIC SIMULATION. MANUAL OR AUTOMATIC GEARS. WET OR DRY CONDITIONS. GREAT GRAPHICS.

CASSETTE CP/M.....328 ONLY / UTILITY /64k RAM-CARD / R 30
REQD.

CASSETTE CP/M USES A 64k RAM-CARD AS A DISK-DRIVE. IT IS LIMITED IN STORAGE CAPACITY, BUT RUNS MUCH FASTER.
YOU CAN NOW DO ANYTHING THAT A NORMAL CP/M SYSTEM DOES USING ONE DISK-DRIVE (BUT YOU CAN'T SWITCH DISKS).
SPECIAL MODULES ALLOW FOR COPYING BETWEEN 'DISK' AND CASSETTE.

CASSETTE INTERFACE....328 ONLY / UTILITY / DISK / R 14
FOR CP/M 2.22

THE DISK CONTAINS TWO PROGRAMS FOR SAVING AND LOADING TO AND FROM CASSETTE WHILST UNDER THE CONTROL OF CP/M. THE CASSETTE CAN PROVIDE THE CP/M USER WITH AN UNLIMITED AMOUNT OF VERY INEXPENSIVE STORAGE SPACE. SOMETHING LIKE A THIRD 'DISK'?

RAMDISK.....328 ONLY / UTILITY /64k RAM-CARD / R 30
REQD.

RAMDISK IS A PROGRAM THAT CONVERTS YOUR 64k RAM-CARD INTO A THIRD DISK-DRIVE. ALTHOUGH IT IS LIMITED TO 64k, IT OPERATES MUCH FASTER THAN A NORMAL DISK-DRIVE. THIS 'DISK-DRIVE' THEN BECOMES DRIVE 'C'.

BUZZ OFF.....MSX ONLY / ARCADE / TOPS / R 35

BERTIE THE BEE HAS A SWEET TOOTH - HE JUST LOVES FRUIT. SO MUCH SO THAT HE IS EVEN PREPARED TO BRAVE THE PERILS OF THE ENCHANTED ORCHARD THE ONLY TROUBLE IS THAT WHEN BERTIE IS ON A BINGE HE FORGETS TO LOOK WHERE HE'S GOING. JUST TO MAKE MATTERS WORSE, EVERY TIME HE EATS A FRUIT, A SEGMENT OF SPIDER'S WEB TAKES IT'S PLACE. CAN YOU GIVE BERTIE A TEN-COURSE BANQUET?

SUPERCHESS.....328 ONLY /BOARDGAME / TOPS / R 40

THIS IMPLEMENTATION OF CHESS HAS SEVEN LEVELS OF PLAY FROM EASY TO DOWNRIGHT DIFFICULT. THE FIRST FOUR LEVELS PLAY WITHIN TOURNAMENT TIME LIMITS, AND LEVEL ZERO RESPONDS IMMEDIATELY. GRAPHIC SCREEN DISPLAY OF BOARD AND PIECES. SIZEABLE LIBRARY OF OPENING MOVES. RECOMMENDED MOVE OPTION IN CASE YOU GET STUCK. ANALYSE MODE, ALLOWS YOU TO SET UP ANY POSITION.
BEATS MOST OTHER COMPUTER CHESS PROGRAMS AND GIVES YOU A CHALLENGING AND INTERESTING GAME OF CHESS.

COMPUTO-LINK.....328/MSX / UTILITY / FANTASTIC / R 51

COMPUTO-LINK ENABLES YOU TO FULLY UTILISE YOUR DAISY BELTEL MODEM BY CONFIGURING YOUR COMPUTER AS A COMMUNICATIONS TERMINAL. ASCII FILES CAN BE TRANSFERRED BETWEEN USERS (OVER THE PHONE) OR YOU CAN SIMPLY TYPE IN TEXT WHICH WILL IMMEDIATELY BE DISPLAYED ON BOTH YOUR TERMINAL AND THE REMOTE TERMINAL.
THE PROGRAM FEATURES AN AUTO-DIALLER WITH 1000 POSSIBLE TELEPHONE NUMBERS. YOU CAN ALSO SPEAK DIRECTLY TO THE REMOTE STATION (TWO-WAY).

1 WHAT ARE THE PARTS OF A COMPUTER?

Now we are going to talk about the kind of computer you might have on your desk or buy to use at home. Whatever model it is, it must have certain parts with clearly-defined functions.

First, there must be some means of telling the computer what you want it to do. We shall call this the 'input device'. Normally, it is a keyboard just like a typewriter's, possibly with some additional function keys for convenience. By means of the input device, we give the computer 'commands', words which it 'understands' as instructions to perform some action. Actually, of course, the machine understands nothing, but it is wired up to respond to certain letter-combinations typed at the keyboard by performing a specific action.

For example, if we type 'PRINT "HELLO" ', the machine will produce the word 'HELLO' either on the screen, on a printer or some such output device.

Which brings us to the second essential component of the computer, one or more 'output devices'. Usually the output device, which is the means by which the user is enabled to see the result of the computer's action, is some sort of screen, either a domestic TV screen or a specially-built monitor, but it could be a printer (rather like the works of a typewriter) or even a plotter, anything which is capable of reproducing text for us to read.

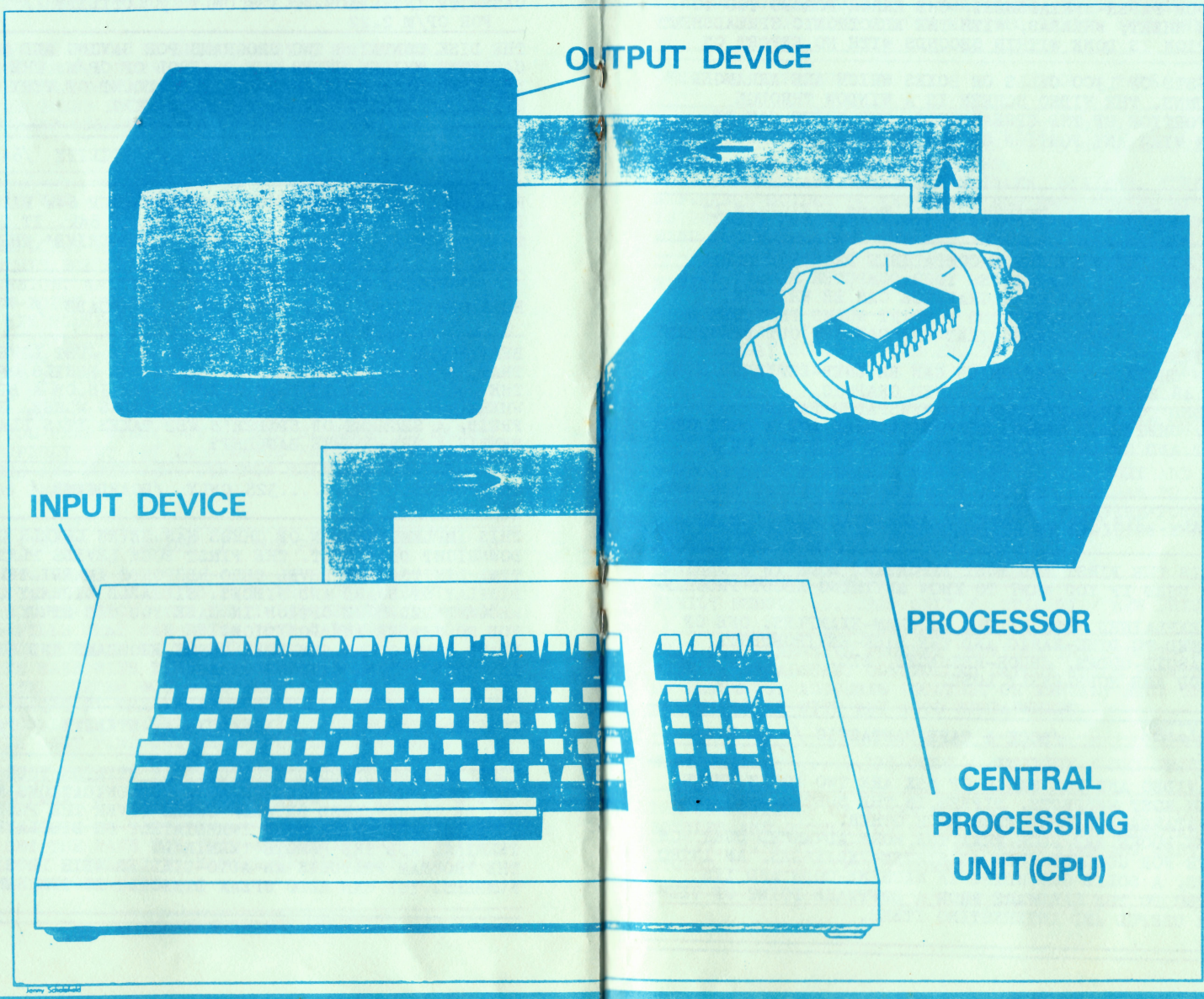
The functions of the input and output devices are easy to understand. It is when we come to the third component, the central processor, that we find ourselves on less firm ground. Yet this is the heart of the computer, the box which does the actual work.

Actually, there is nothing mysterious about the central processor, although its actual design may be a bit complicated. It is little more than a collection of electronic switches connected together so that, in certain circumstances, they can switch each other on and off.

By general agreement, various combinations of 'on' and 'off' switches are recognised as codes representing numbers and letters and the central processor uses these codes to perform all its actions.

There is one chip (an array of microscopic switches encapsulated into a single component) which is designed to perform arithmetic functions on numbers fed into its storage 'registers'. This is the heart of the computer, known as the CPU or Central Processing Unit. It is served by other chips which store numbers or move them around inside the machine.

These three components are the essential parts of a computer. Usually, certain other components or 'peripherals' are added to the system, such as 'storage devices' in the form of 'disk or tape drives' (same principle as a domestic cassette recorder), but these are not essential to the operation of the machine. ■



Jenny Scholefield

The conventional codes used to represent the alphabet

There are usually eight electronic switches forming a 'register' to hold a code. One is used for additional control signals. The remaining seven can take 128 different on/off patterns. If you imagine 0 representing an 'off' switch and 1 representing an 'on' switch, the patterns or codes for the letters of the alphabet appear as follows:

A	00100001
B	00100010
C	00100011
D	00100100
E	00100101
F	00100110
G	00100111
H	00101000
I	00101001
J	00101010
K	00101011
L	00101100
M	00101101
N	00101110
O	00101111
P	00110000
Q	00110001
R	00110010
S	00110011
T	00110100
U	00110101
V	00110110
W	00110111
X	00111000
Y	00111001
Z	00111010

SPREAD-SHEET.....328/MSX / UTILITY / TOPS / R 47

AN ELECTRONIC SPREADSHEET PROGRAM CREATES A SPECIAL ENVIRONMENT WITHIN THE COMPUTER SO THAT REPETITIVE CALCULATIONS CAN BE PERFORMED EASILY AND SIMPLY. IF YOU TAKE A PIECE OF PAPER AND ON IT, PREPARE EITHER AN INCOME STATEMENT, A CHEQUE-BOOK BALANCE OR AN AVERAGE OF YOUR EXAMINATION RESULTS; THEN YOU HAVE PREPARED A SPREAD-SHEET. WITH THE ELECTRONIC SPREADSHEET, THE COMPUTER DOES ALL THE CALCULATION AND YOU SIMPLY KEY IN THE VALUES AND THE FORMULAE FOR THOSE CALCULATIONS. WITH YOUR PAPER-SPREADSHEET, ONE ERROR FORCES YOU TO RECALCULATE THE WHOLE SHEET; WHEREAS, WITH THE ELECTRONIC SPREADSHEET THE ENTIRE RECALCULATION IS DONE WITHIN SECONDS WITH NO EFFORT ON YOUR PART.
THE SPREADSHEET CONSISTS OF 1300 CELLS OR BOXES WHICH ARE ARRANGED IN 50 ROWS OF 26 COLUMNS. THE VIDEO SCREEN IS A WINDOW THROUGH WHICH YOU CAN SEE A PORTION OF THE SPREADSHEET. THE SHEET CAN BE MOVED SO THAT YOU CAN VIEW ANY PORTION OF THE SPREADSHEET THROUGH THE WINDOW.

PRINT WORKS.....328/MSX / UTILITY / TOPS / R 47

N.B. THIS PROGRAM IS FOR USE WITH EPSON-COMPATIBLE PRINTERS ONLY.
PRINT-WORKS IS A PROGRAM WHICH CONFIGURES YOUR COMPUTER MEMORY INTO FIVE PAGES ON WHICH TEXT AND CHARACTER GRAPHICS CAN BE WRITTEN.
A PAGE IS 80 COLUMNS WIDE BY 81 ROWS LONG - PRINT WORKS PROVIDES A FULL SCROLLING EDITOR FOR EACH PAGE SO THAT YOU CAN SET OUT THE TEXT AND GRAPHICS AS REQUIRED.
EACH PAGE HAS ITS OWN CHARACTER FONT WHICH CAN BE USER-DEFINED. EACH PAGE CAN ALSO HAVE A 48 BY 64 DOT USER DEFINED GRAPHIC.
PRINT WORKS CAN BE USED TO PREPARE YOUR OWN NEWSPAPER PAGES, ADVERTISING LEAFLETS, CHRISTMAS CARDS, BIRTHDAY CARDS AND A HOST OF OTHER USES WHERE TEXT AND GRAPHICS ARE REQUIRED TO BE MIXED ON A PAGE.

BASIC REFERENCE MANUAL...318/328 / BOOK / ESSENTIAL / R 38

THIS ESSENTIAL BOOK IS THE FIRST AND MOST IMPORTANT BOOK IN A SERIES WHICH SIMPLY MUST BE USED IF YOU WANT TO KNOW ANYTHING ABOUT PROGRAMMING.
ALL BASIC WORDS ARE EXPLAINED WITH LITTLE PROGRAM-EXAMPLES. ONE OF THE CHAPTERS IS DEVOTED TO DISK-BASIC AND THE DISK-ENVIRONMENT.
THERE ARE TABLES OF ASCII-CODES, ERROR-MESSAGES, ETC.
WITHOUT THIS BOOK, YOU ARE HOPELESSLY LOST.

THE MAGIC OF SPECTRAVIDEO / MSX /BOOK & TAPE/FANTASTIC / R 43

THE MAGIC OF SPECTRAVIDEO AND THE MAGIC OF MSX ARE TWO BOOKS WHICH ALTHOUGH IDENTICAL IN MOST RESPECTS, DIFFER IN THE SUBTLE WAYS THAT THE 328 AND THE MSX-STANDARD DIFFER FROM EACH OTHER.
THE CONTENTS OF THESE BOOKS ARE SUCH THAT THE MORE ADVANCED USER NOW HAS A HANDY REFERENCE FOR GOODIES SUCH AS SYSTEM-VARIABLES, AN INTRO TO THE NUMBER-SYSTEMS, A SOLID GROUNDING IN MACHINE-CODE AND ITS APPLICATIONS, AN INTRO TO THE HARDWARE FROM A SOFTWARE POINT OF VIEW AND MANY, MANY OTHER USEFUL AND INTERESTING ITEMS.

PRACTICAL PROGRAMS FOR THE SV.328 /BOOK & TAPE/ TOPS / R 32

THIS BOOK IS THE FIRST TRUE APPLICATIONS OF MACHINE-CODE GAMES ROUTINES AND OTHER INTERESTING ROUTINES; TO BE WRITTEN. IT IS ALSO THE FIRST IN A SERIES (OF 2 OR MORE).
STARTING YOU OFF IN THE FAMILIAR REALM OF BASIC, YOU ARE SLOWLY BUT SURELY MOVED OVER TO THE EQUIVALENT IN MACHINE-CODE WHERE ALL THE NECESSARY SUB-ROUTINES ESSENTIAL TO THE FORMATION OF A FULL GRAPHIC GAME ARE GIVEN AND EXPLAINED.
THE BOOK CULMINATES IN A FULL GAME, USING THOSE SUB-ROUTINES AND PROVIDING YOU WITH ENOUGH MEAT TO DESIGN YOUR OWN GREAT GAMES.

-MSX- AN INTRODUCTION /BOOK & TAPE/ TOPS / R 30

THE BOOK TAKES YOU FROM THE SIMPLEST CONCEPTS OF MSX BASIC TO THE COMPLEX TECHNIQUES OF USING ITS STUNNING GRAPHICS AND MUSIC CAPABILITIES; THE MSX MACRO-LANGUAGES.
WITH DRAWING AND PAINTING PROGRAMS, SPRITES, TUNES TO PLAY, EASY REFERENCE GUIDES TO KEY WORDS AND MANY USEFUL HINTS AND TIPS, THIS IS PERFECT FOR BOTH BEGINNER AND ADVANCED PROGRAMMERS WISHING TO FIND OUT MORE ABOUT THE REVOLUTIONARY MSX SYSTEM.

NINJA.....328/MSX / ARCADE / TOP PROGRAM / R 20

THE ANCIENT NINJA WARRIORS OF JAPAN ARE FIGHTING OUT EPIC BATTLES ON THE DISPLAY SCREEN. THIS ARCADE PRIZE-WINNER FEATURES AMAZING GRAPHICS AND IS ONE OF THE 'ORIGINALS' OF THE ARCADE.

TURBOAT.....328 ONLY / ARCADE / TOP PROGRAM / R 20

TURBOAT MUST PENETRATE THE ENEMY'S SIDE PORT AND DESTROY THE JETFOILS WITH ITS UNLIMITED SUPPLY OF BULLETS. IT HAS TO PICK UP THE ENEMY'S SECRET RADAR FOR ANALYSIS BY LOSING ITS FIRE-POWER.
THE FINAL ZONE IS TURBOATS ESCAPE BY AVOIDING THE ENEMY GUIDED MISSILE AND TORPEDO.

FAMILY BUDGET.....328 ONLY / UTILITY / NEW PRICE / R 15

THIS COMPREHENSIVE PROGRAM PROVIDES THE BUDGET-CONSCIOUS HOUSEHOLDER WITH A SPREADSHEET CONTROLLED LAYOUT WHICH PROVIDES FOR ANY NUMBER OF PAYMENTS, ACCOUNTS (MONTHLY OR ANNUAL) ETC. UPDATING OF TOTALS IS AUTOMATIC AS WITH ANY GOOD SPREADSHEET.

WINE CELLAR .....328 ONLY / UTILITY / NEW PRICE / R 12

THE CONNOISSEUR OF WINES NOW HAS AVAILABLE A STOCKHOLDER FOR THOSE BOTTLES AND BOTTLES OF SOMETIMES OBSCURE DELECTABLES CALLED WINE. ESTATE, VINTATE, ETC., AND RE-ORDER TIME ARE PART OF THE INFORMATION STORED IN THIS NIFTY PROGRAM.

ANIMATE.....328 ONLY / UTILITY / TOP PRGRAM / R 15

IF YOU LIKE ANIMATION BOTH IN TWO AND THREE DIMENSIONS, THIS PROGRAM PROVIDES YOU WITH ONE OF THE MOST USEFUL MOTION-PICTURE AIDS EVER. ANIMATE ALLOWS YOU TO DRAW, MANIPULATE AND ANIMATE TWO AND THREE DIMENSIONAL SHAPES. THESE CAN BE SAVED TO OR LOADED FROM TAPE; SAVED DIRECTLY TO VIDEO-MACHINE; CREATED, EDITTED, DISPLAYED AND ANIMATED IN 4, 6, 8, 12, 16, 24, OR 48 FRAMES OF CORRESPONDINGLY LARGE TO SMALL FRAME SIZES.
DEMO FILES ARE PROVIDED ON THE TAPE.

MUSIC MASTER.....328 ONLY / UTILITY / TOP PROGRAM / R 12

THE ASPIRING YOUNG MUSICIAN CAN ENTER MUSICAL NOTES DIRECTLY FROM THE SHEET-MUSIC WITH THIS PROGRAM. THE MENU PROVIDES FOR 'COUNT', 'VOICE' 'REST', 'TEMPO', 'SOUND', 'EDIT', 'STORE', 'PLAY', 'TAPE' AND 'CLEAR' UP TO 10 PAGES OF 240 NOTES EACH CAN BE STORED AND SELECTED PAGES CAN THEN BE REPLAYED AS DESIRED.

GUNFRIGHT.....MSX ONLY / ARCADE / NEW RELEASE / R 40

HOW'DE PARDNERS
NOW Y'ALL LISTEN UP. SEEMS WE GOT OURSELVES A NEW SHERIFF IN TOWN. RECKON HE'S GONNA CLEAR THE TOWN OF THE MEANIST AND FASTEST GUN TOTIN BUNCH OF ROOTIN' TOOTIN' GUN SLINGERS WHICH EVER DID HIT THE WILD WEST; GOES BY THE NAME OF SHERIFF QUICKDRAW. YES SIREE.
SHERIFF QUICKDRAW SPRINGS INTO ACTION AND WITHIN THE BLINK OF AN EYE, IS AWAY ON HIS TRUSTY STEED, PANTO.

BLAGGER.....328/MSX / ARCADE / NEW RELEASE / R40

FOLLOW THE EXPLOITS OF ROGER THE DODGER, MASTER BURGLAR, AS HE WORKS HIS WAY THROUGH NUMEROUS BUILDINGS - SEARCHING FOR THE GOLDEN KEYS, ROBBING SAFE AFTER SAFE BUT ALWAYS ON THE ALERT TO THE MANY OBSTACLES AND ALARM SYSTEMS AND THOSE SPOOKY, COOKY NIGHTWATCHMEN.
WHEN ONE SCREEN IS CLEANED OUT YOU MOVE TO THE NEXT LEVEL IN AN ENDLESS PATTERN OF BANKS, SHOPS AND HOUSES.

EDDIE KIDD JUMP.....328/MSX /SIMULATION/ NEW RELEASE / R20
CHALLENGE

IF YOU'VE EVER WONDERED HOW IT FEELS TO BE EDDIE KIDD...DEFYING DANGER AND GRAVITY, SOARING ON A MOTOR-BIKE OVER A LONG LINE OF CARS...HERE'S YOUR CHANCE TO FIND OUT.
JUST LIKE EDDIE, YOU'LL START BY TRYING TO CLEAR BARRELS ON A BICYCLE...GRADUALLY WORKING UP TO ATTEMPTING EDDIE'S DARING LEAPS WHICH HAVE THRILLED MILLIONS AROUND THE WORLD.

JACK THE NIPPER.....MSX ONLY / STRATEGY / NEW RELEASE / R 45

HE'S WICKED. HE'S EVIL. CAN HE EVER BE STOPPED? JACK THE NIPPER PLAYS THE TRICKS AND CREATES THE HAVOC WE ALL WOULD HAVE LIKED TO HAVE ENJOYED. SO WHY WAIT ANY LONGER???

Boolean Algebra for Computer Logic

Table 1-6. Rules for Binary Subtraction

A	−	B	−	Previous Borrow	=	Difference	New Borrow
0	−	0	−	0		0	0
0	−	0	−	1		1	1
0	−	1	−	0		1	1
0	−	1	−	1		0	1
1	−	0	−	0		1	0
1	−	0	−	1		0	0
1	−	1	−	0		0	0
1	−	1	−	1		1	1

Here is another example:

$$\begin{array}{rl} 0110 & \text{(decimal 6)} \\ -0011 & \text{(decimal 3)} \\ \hline 0011 & \text{(decimal 3)} \end{array}$$

1st digit: 0−1 = difference of 1 and borrow 1.
2nd digit: 1 borrowed from top row (becomes 0) and 0−1 is a difference of 1 and borrow 1.
3rd digit: 1 borrowed from top row (becomes 0) and 0−0 = 0.
4th digit: 0−0 = 0.

Table 1-6 tabulates the rules for binary subtraction.

When we progress to the "fascinating complement" (next section) you will find binary subtraction much simplified.

1-5. HANDLING POSITIVE AND NEGATIVE BINARIES

You have positive and negative values in binary notation just as in base 10 arithmetic. It is time now to consider the binary scheme of manipulating either polarity in calculations. You will find binary subtraction by "complements" much simpler than the straight arithmetical method described above.

The Fascinating Complement

A *complement* is that quantity or amount which, when added to a given quantity, completes a whole. For example, in Fig. 1-5,

if you want to complete an angle of 90°, the arc BD is the *complement* of arc AB, and the angle BCD is the complement of the angle ACB. Another way of saying this is that, for an angle of 90° as a whole, the complement of 30° is 90°−30° = 60°.

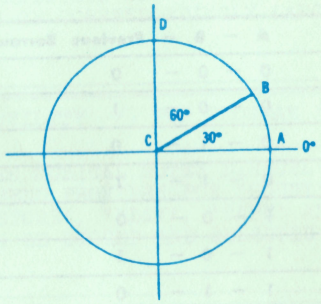


Fig. 1-5. Complement of an angle.

For decimal numbers the complement of 460 is 1000−460 = 540. The complement of 25 is 100−25 = 75. Note in each case that the arithmetical complement is actually the *difference* between a number and the power of the base next in series. When you complement a positive number, it becomes a *negative* value, and the complement is a *positive* number. Synonyms for the "complement" are: inversion, reversal, the opposite, the reverse, the inverse, the converse.

In conventional arithmetic, suppose you subtract 235 from 485:

```
 485  (minuend)
-235  (subtrahend)
 250  (answer)
```

You can do the same thing by adding rather than subtracting if you complement the subtrahend and continue as in normal addition:

```
 485  (minuend)
+765  (complement of subtrahend)
1250  (answer with 1 overflow)
```

In complementary arithmetic, the most significant digit is not a part of the answer numerically (1 in the above example) and the answer (250) remains.

Let's see what happens if you have a result which is a negative number rather than a positive number. Suppose you have 235−485. Normally you simply subtract the smaller number from the larger number and affix the sign of the larger number to the answer. Hence −485 + 235 = −250.

In complementary arithmetic:

```
 235  (minuend)
+515  (complement of a negative number is positive)
 750  (complement of 250)
```

Hence recomplementing 750 gives −250 (answer). The complement of a complement is the original number.

Simply note from this example that when a resultant is a negative number, it will be indicated by the appearance of the complement form in the answer. The answer is then recomplemented to get the correct negative value.

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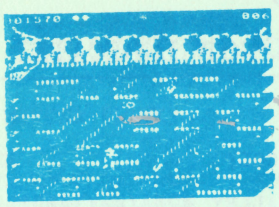
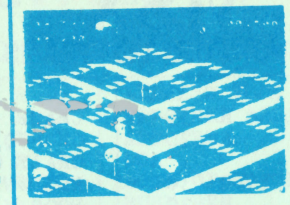


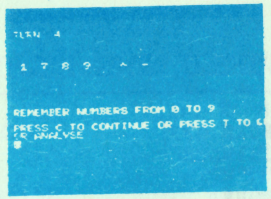
211T OLD MAC FARMER

Help Old Mac Farmer clear his field of pesty bugs by destroying their eggs with your limited supply of insecticide. Then, protect the crops from the constant attack of giant birds and nasty ducks.



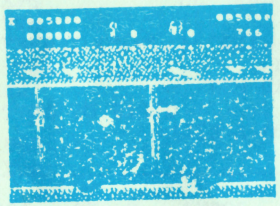
216T TETRA HORROR

Face fierece skulls and ghosts and avoid rolling boulders and deadly pits in your role as TAMO the adventure. Use your skill and fire gun to destroy your enemies, while battling your way through the TETRA CASTLE to obtain treasure.



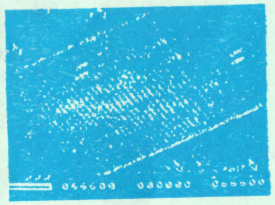
667T SPECTRABRAIN

The computer is thinking of a number. A number that could be in the tens, the hundreds, the thousands or even, the tens of thousands--can you guess it? There's no end to the fun and challenge of SPECTRABRAIN.



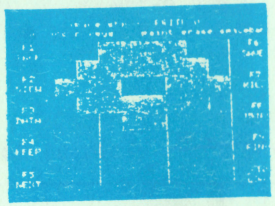
292T NINJA

The ancient Ninja warriors of Japan fight it out in this epic battle for fame and fortune.



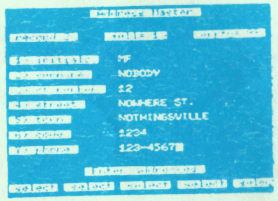
242T TURBOAT

Your Turboat must avoid missles and torpedo in a mission to penetrate enemy forts, destroy jetfoils and pickup secret radars.



602D SPRITE MASTER

SPRITE MASTER is a handy utility for the would-be game or educational software developer. This powerful program lets you create and edit 8X8 or 16X16 bit sprites; then, integrate them in your own BASIC language programs.

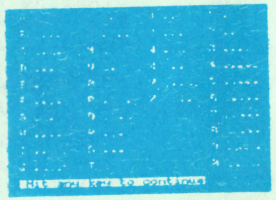


603D ADDRESS MASTER

ADDRESS MASTER lets you maintain and organize addresses of friends, acquaintances, business associates, and the like. You can create files, edit entries, sort addresses, search for an address, and even, print labels with this useful program.

617T GRAPH MASTER

GRAPHMASTER turns your computer into a graphing machine. You can perform experiments in mathematics and geometry that will help you learn concepts taught in secondary school.



654T MORSE CODE TUTOR

If you have a desire to learn morse code, this is the program for you. With a few hours on MORSE CODE TUTOR, even the raw beginner will be ready to tap-out his own coded messages.

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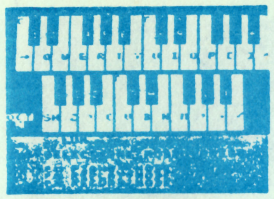
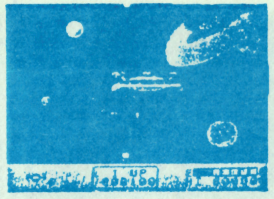


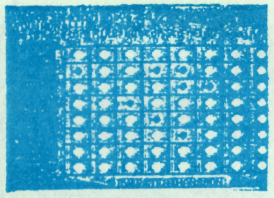
236C MUSIC MENTOR

Turn your computer into a musical instrument and create, edit and play your own synthesised music.



237C SUPER CROSS FORCE

Master the dual countersync laser--firing from two independent vantage points simultaneously--to out maneuver formidable fleets of "Morpul" ships and save the galaxy.

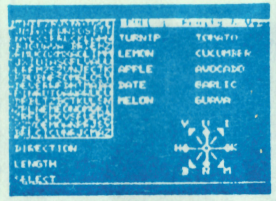


238T OTHELLO

The computer offers a clever and intelligent opponent in this classic game of strategy--Othello.

235T INTRODUCTION TO BASIC

Introduction to BASIC reveals the secrets of advanced BASIC programming enabling you to take full command of the Spectro's powerful graphics and sound capabilities.

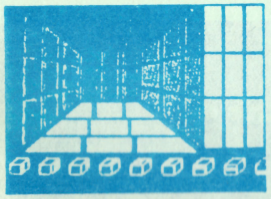


252T FINDIT

Develop your vocabulary and practice your spelling with this automated derivative of the classic crossword puzzle.

234T SPECTRON

Weather wave after wave of alien invaders to win the galactic wars and save your beloved motherland in this classic arcade game.

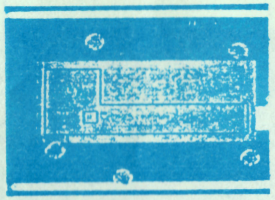


245T CRYPTIC CUBE

A computerized version of the popular rubic cube puzzle.

257T RESCUE

Avoid shells and radio aerials while fly over jagged peaks in your helicopter to save the three brave soldiers.



260T MINI GOLF

Turns your computer into a 9 hole miniature golf course with enough challenge for even a golf pro.

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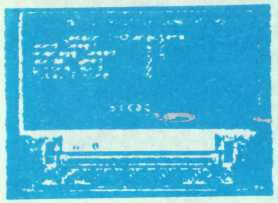
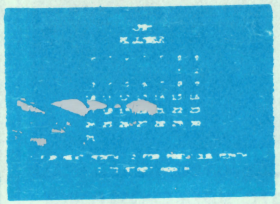


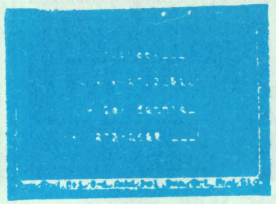
288T SPECTRA TYPE

Spectra Type is a fun way to improve your typing speed and accuracy.



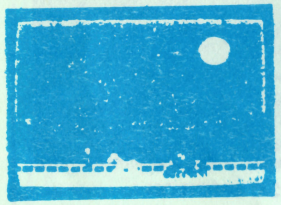
322T CALENDAR

Calendar, opens a new window to the future and past, enabling you to view a calendar of any month in the 19th, 20th, and 21st centuries.



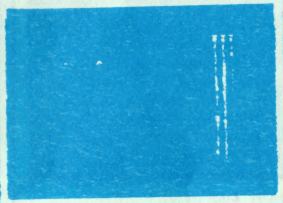
275T/D SPECTRA FILE CABINET

File Cabinet provides a quick and easy way to save and organise your valuable home information.



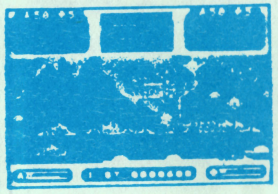
326T HORSE RACE

Bet on the races then cheer your horse to victory as they race for the post, right before your eyes.



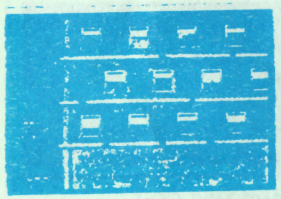
296T SMASH

Bash the ball through layer after layer of bricks to score points and break through the wall to win.



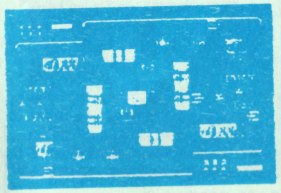
SECTOR ALPHA SD220C

Defend Sector Alpha against robot invaders from the Throgg Empire. Combine your careful aim with a periscope that can rotate a full 360 degrees and a powerful laser to defend every parameter.



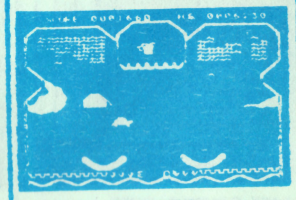
FRANTIC FREDDY SC232C

Your cat is caught in a flaming inferno: you must help Frantic Freddy, the fearless fireman, put out the flames and save the day.



AMOURED ASSAULT SD233C/T

Dual man against man or man against machine in this exciting and challenging World War II tank battle.



FLIPPER SLIPPER SD291C

Turn your computer into a pinball machine -- beat the highest score and become a video pinball wizard.

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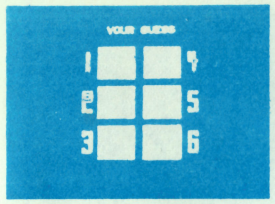
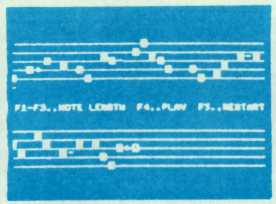


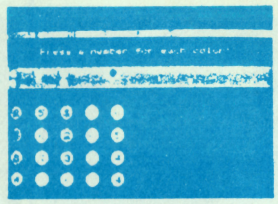
315T MEMORY TRAINER

How good is your recall? Find out with memory trainer.



316T MUSIC COMPOSER

An easy way to play and compose your own tunes.

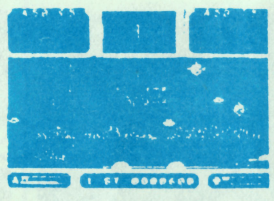


335T SPECTRAMIND

Find the hidden color sequence within 8 attempts.

219T NEW YORK BOMB BLITZ

Your engines are disabled as your ship enters the earths' atmosphere. Falling rapid towards New York city, you must clear a landing site or be devastated by the towering sky scrappers.



220C SECTOR ALPHA

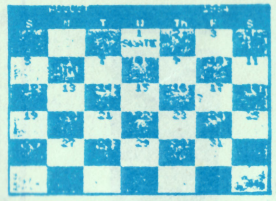
Defend Sector Alpha against robot invaders from the Throgg Empire. Combine your careful aim with a periscope that can rotate a full 360 degrees and a powerful laser to defend every parameter.

255T NOMIS

Match wits with "Nomis" the master computer brian - test and develop your memory in this challenging number game.

227T/D SPECTRA CHECKBOOK

This program offers a friendly and convenient tool for maintaining and updating your bank balance.



228T SPECTRA DIARY

Plan and detail your appointments, errands and important days to remember with this automated diary.

274T BONE OF CONTENTION

A game that will challenge and test your elementary math skills.

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